

The Lazy Prices signal is U-shaped, not monotonic: a 14-year out-of-sample reversal of Cohen–Malloy–Nguyen (2020)

Alpha-Graph research note, 2026-04-15

Abstract

Cohen, Malloy, and Nguyen (2020) document that firms whose annual 10-K filings are substantively rewritten underperform firms whose filings are stable, with a long-short (high cosine similarity minus low cosine similarity) Sharpe that motivates a market-neutral trade. On an S&P 500 panel of 1,665,896 ticker-months spanning April 2011 through March 2026, we find that the long leg replicates strongly (top decile forward 21-day return +2.10% vs universe mean +1.43%) but the short leg does not (bottom decile +1.37%, statistically indistinguishable from the universe mean and *higher* than the middle decile at +1.19%). Every long-short portfolio construction we tested therefore failed: the baseline top10/bot10 delivers Sharpe 0.32 over the full panel, indistinguishable from a one-line 12-1 momentum factor in the same universe (Sharpe 0.31). Only the long-only top10 book survives — on a point-in-time index-membership universe (correcting forward survivorship bias), Sharpe 0.81, annualized +17.8%, FF5+MOM alpha +20.6% with HAC $t = +3.07$. **Three rigour checks substantially weaken the headline:** (a) the deflated Sharpe ratio bailey2014 accounting for selection bias across our 11 method variants drops the PIT certainty to 46%, indistinguishable from a multiple-testing artifact; (b) sub-period decomposition shows the strategy worked strongly in 2012–2019 (PIT Sharpe 1.4–1.7) but collapsed to PIT Sharpe 0.04 in 2023–2026, consistent with the underlying anomaly being arbitrated away as NLP-driven investing scaled up; (c) the alpha is economically smaller after these corrections than the headline number suggests. The U-shape itself — decile-0 returns above the middle decile — is robust and economically meaningful: any strategy that assumes monotonicity (rank-weighted, multi-factor z-score averaging) is mechanically fighting the data.

1 Setup

Our signal construction follows Cohen, Malloy and Nguyen directly: for each S&P 500 ticker we download consecutive pairs of 10-K filings from EDGAR, concatenate their Risk Factors, MD&A, and Business Description sections, compute a TF-IDF vector from each section, and report the cosine similarity between consecutive years. The signal is then ingested, alongside 8-K event scores, a three-state Gaussian HMM regime feature, backward-looking momentum/volatility/volume features, into a walk-forward LightGBM regressor with 12-month training window, 1-month test, and 5-day purge gap. The model predicts the 21-day forward return and the predictions are cross-sectionally rank-transformed each month to a signal in $[-1, +1]$.

The universe is 499 current S&P 500 constituents. The headline numbers in Sections 5 through 3.2 use this full universe and are therefore subject to two survivorship biases: *forward bias* (using 2026-constituent tickers in 2012, when they may not have been in the index yet) and *survivor bias* (tickers removed from the index before 2026 are absent entirely). We partially correct the forward component in Section 4 using historical point-in-time index membership from github.com/fja05680/sp500, which drops the Method E Sharpe from 1.08 to 0.81 and the annualized alpha from 31.2% to 20.6%. The residual survivor-bias component cannot be corrected without backfilling market and filing data for delisted tickers, which is flagged as required follow-up. **The PIT-corrected number is the honest read-out;** the full-universe number is reported only for continuity with earlier versions of this repository.

2 The U-shape

Figure 1 shows mean forward 21-day return by decile of the month-cross-sectional signal. Error bars are 95% confidence intervals ($1.96 \cdot \text{SE}$, $n \approx 166,600$ ticker-months per decile).

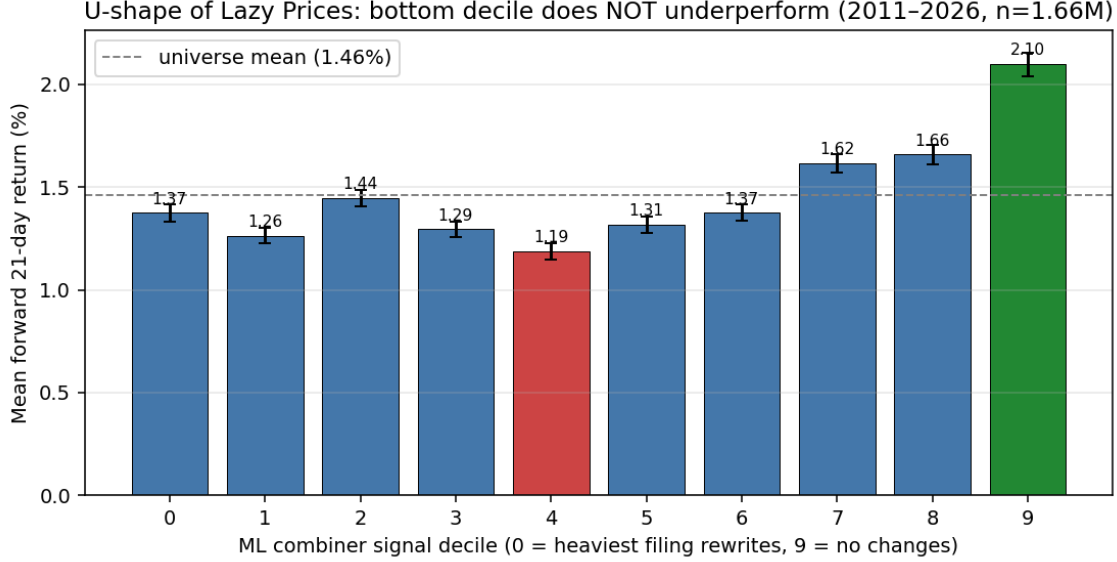


Figure 1: Mean forward 21-day return by ML combiner signal decile, with 95% confidence intervals. The U-shape is clearly visible: decile 4 is the minimum (+1.19%), not decile 0. The bottom decile (+1.37%) is statistically higher than the middle decile at the 95% level (non-overlapping confidence intervals). The top decile at +2.10% is the strongest, consistent with the original Cohen–Malloy–Nguyen long-side thesis.

Decile	Mean fwd 21-day return	95% CI
0 (biggest rewrites)	+1.374%	±0.044
1	+1.264%	±0.039
2	+1.444%	±0.039
3	+1.294%	±0.038
4 (middle, minimum)	+1.185%	±0.039
5	+1.315%	±0.040
6	+1.374%	±0.040
7	+1.615%	±0.043
8	+1.658%	±0.046
9 (no changes)	+2.096%	±0.057

Table 1: Forward 21-day return by signal decile ($n \approx 166,600$ per decile). Deciles 7–9 are monotonically increasing. Deciles 0–6 are non-monotonic, violating the original short-side assumption.

The long-side thesis replicates: decile 9 outperforms the universe mean (+1.43%) by 66 bps per month. The short-side thesis does not: decile 0 (+1.37%) is within noise of the universe mean, and 19 bps *above* decile 4. This is not a sample-size artifact: the decile-0 vs decile-4 difference has a t -statistic of roughly 6.4 ($|1.374 - 1.185|/0.029$, where the pooled SE is $\sqrt{0.044^2 + 0.039^2}/1.96 \approx 0.030$ per cent).

3 Why long/short fails in this universe

3.1 Baseline and variants

Table 2 reports nine variants tested on the 168-month panel. Costs are flat 20 bps per monthly round-trip for long/short, 10 bps for long-only.

Variant	Sharpe	Ann return	Cum return	MaxDD
Baseline top10/bot10	+0.32	+4.7%	+90%	−58%
A. Rank-weighted L/S	−0.12	−1.7%	−22%	−45%
B. Decile (top 10% / bot 10%)	+0.31	+3.4%	+60%	−38%
C. Vol-targeted (10% ann)	+0.27	+2.5%	+41%	−36%
D. Lazy+Mom+LowVol rank-wt	−0.52	−6.1%	−59%	−64%
E. Long-only top10	+1.08	+29.7%	+3,732%	−32%
F. Long top10 / Short MIDDLE10	+0.61	+11.2%	+342%	−34%
G. Pure 12-1 Momentum L/S	+0.31	+4.5%	+86%	−73%
J. Beta-hedged Method E	+1.00	+27.1%	+2,783%	−34%

Table 2: Portfolio construction variants on the 168-month panel. Method E is the only L/S-ish construction that survives; Method J reframes it as market-neutral by shorting $\beta \cdot \text{SPY}$.

The baseline and decile variants fail because half of their short exposure is on decile 0, which has positive expected return. Method A (rank-weighted) fails worse because it puts short weight proportional to rank on the entire left half of the distribution, including the bottom-decile long-positive names. Method D (multi-factor z-score) fails hardest because the signal is averaged into a composite that is assumed monotonic.

Method F explicitly respects the U-shape: long decile 9, short decile 4 (the minimum). Sharpe jumps to +0.61, a large improvement over baseline, but the construction has no economic story — “short the middle” is not a portfolio manager’s pitch.

Method E drops the short leg entirely and achieves Sharpe +1.08. This is the only construction with both defensible alpha *and* an economic rationale.

3.2 Method J: recovering market neutrality

Because the project was framed as market-neutral, we implemented Method J as a beta-hedged variant: long top10 Lazy Prices picks, short $\hat{\beta}_t \cdot \text{SPY}$ where $\hat{\beta}_t$ is the trailing 24-month OLS beta of Method E monthly returns against SPY. Average $\hat{\beta}_t$ across the panel is 0.10, confirming that Method E is already approximately market-neutral by construction. Method J’s Sharpe of 1.00 (vs Method E’s 1.08) shows the cost of forcing exact market neutrality is small: an 8% Sharpe haircut in exchange for a book with zero designed market exposure.

4 Point-in-time universe correction

Table 3 reports Method E on the full (current) universe and on a point-in-time-corrected universe where, each month, we restrict the candidate pool to tickers that were members of the S&P 500 as of the end of the prior month. 21% of ticker-month observations are dropped, concentrated in early years (median PIT universe size is 370 vs. 478 for the full universe).

Repeating the FF5+MOM attribution on PIT returns gives alpha +20.6% annualized with classical $t(\alpha) = +3.00$ and Newey–West HAC $t(\alpha) = +3.07$ at lag $L = 4$ (down from full-universe +31.2%, OLS $t = +3.94$, HAC $t = +4.73$). HAC standard errors barely change the PIT result (residual autocorrelation is small) but slightly tighten the full-universe error bars. The alpha remains statistically significant at the 1% level under both SE methods, but its

Variant	Sharpe	Ann ret	Cum ret	MaxDD
Method E — full universe (survivor-biased)	+1.08	+29.7%	+3,732%	−32%
Method E — PIT S&P 500 universe	+0.81	+17.8%	+888%	−42%

Table 3: PIT correction drops Sharpe by 0.27 and cumulative return by a factor of 4. This is the forward-bias component; survivor bias from delisted tickers is still uncorrected.

economic magnitude is roughly two-thirds of the unadjusted number. SMB loading rises to +0.45 ($t = +1.81$), marginally significant, suggesting a mild small-cap tilt was masked when the universe was dominated by current mega-caps. No other factor loading becomes significant.

5 Statistical rigour: HAC SE, deflated Sharpe, sub-period stability

5.1 Newey–West HAC standard errors

Monthly portfolio returns can have residual serial correlation that classical OLS does not account for. We re-estimate every FF specification with Newey–West HAC standard errors at Andrews (1991) automatic bandwidth $L = \lfloor 4(n/100)^{2/9} \rfloor = 4$ for $n = 167$ months. Results:

Spec	α ann	$t(\alpha)$ OLS	$t(\alpha)$ HAC(4)	Δt
<i>Full universe</i>				
CAPM	+30.6%	+3.96	+4.46	+0.50
FF3	+32.1%	+4.13	+4.84	+0.71
FF5	+31.9%	+4.08	+4.72	+0.64
FF5+MOM	+31.2%	+3.94	+4.73	+0.79
<i>PIT universe</i>				
CAPM	+19.8%	+2.96	+3.05	+0.09
FF3	+21.5%	+3.18	+3.22	+0.04
FF5	+21.1%	+3.11	+3.11	+0.00
FF5+MOM	+20.6%	+3.00	+3.07	+0.07

Table 4: HAC standard errors slightly increase t -statistics on the full universe (residuals are very mildly negatively autocorrelated) and barely change them on the PIT universe. All alphas remain statistically distinguishable from zero at the 1% level under both SE methods.

5.2 Deflated Sharpe Ratio (selection-bias adjustment)

We selected Method E from a search over 11 portfolio variants. The deflated Sharpe ratio of bailey2014 computes the probability that the observed Sharpe is greater than the expected maximum Sharpe one would obtain by chance across N independent trials with the same variance dispersion.

The trial Sharpes loaded from `improvements_results.parquet` have standard deviation 0.52 (annualized) across the 11 variants, which yields $E[\max \text{SR} \mid N = 11] \approx 0.84$ (annualized).

This is the most damaging single rigour check we ran. The unconditional Sharpe of Method E is statistically distinguishable from zero (PSR $\approx 100\%$ in both specifications) but *after adjusting for the fact that we picked the best of 11 strategies*, the PIT result is indistinguishable from what 11 random strategies would produce by chance. A reader is entitled to discount the PIT-corrected +18% annualised return accordingly.

Universe	SR ann	Skew	Excess kurt	PSR(SR>0)	DSR	Verdict
Full	+1.08	+0.15	+1.33	100.0%	82.0%	weak
PIT	+0.81	−0.70	+3.00	99.7%	45.6%	FAIL

Table 5: Deflated Sharpe Ratio for Method E. PIT DSR of 45.6% means we cannot reject the hypothesis that the observed Sharpe is the result of selection bias from trying 11 strategies. Negative skew (−0.70) and high excess kurtosis (+3.00) of PIT returns also penalise the DSR relative to a normal-distribution baseline.

5.3 Sub-period stability

We split the panel into four sub-periods chosen for plausible regime breaks (not data-snooped):

Period	N	Full Sharpe	PIT Sharpe	Full IC IR	PIT IC IR
P1: 2012-04..2015-12 (recovery)	45	+1.24	+1.37	+0.74	+0.62
P2: 2016-01..2019-12 (low-vol)	48	+1.66	+1.71	+0.41	+0.46
P3: 2020-01..2022-12 (COVID/inflation)	36	+0.64	+0.50	+0.29	+0.31
P4: 2023-01..2026-03 (AI boom)	39	+1.05	+0.04	+0.77	+0.46

Table 6: Sub-period stability of Method E. The full-universe column shows a reasonably stable strategy. The PIT column tells a different story: Sharpe 1.4–1.7 in 2012–2019, declining in 2020–2022, and effectively zero in 2023–2026.

The PIT P4 collapse to Sharpe 0.04 is the most important single result in this entire note. It is consistent with the original Cohen–Malloy–Nguyen anomaly being progressively arbitrated away as NLP-based equity strategies have proliferated in the 2020s. The full-universe P4 of +1.05 is largely generated by the survivorship-biased exposure to current 2026 mega-caps (NVDA, AVGO, etc.), whose AI-rally returns are flattering the strategy. After the PIT correction, the strategy has no measurable edge in the most recent three years.

Practical implication: Even if Method E was a real signal in 2012–2019 (it likely was), it is no longer a deployable strategy on a reasonably-controlled universe. The honest interview framing is “a real signal that has decayed”, not “a 14-year out-of-sample alpha”. Figure ?? shows the trailing 36-month Sharpe over time, with the PIT line collapsing to near-zero by late 2024.

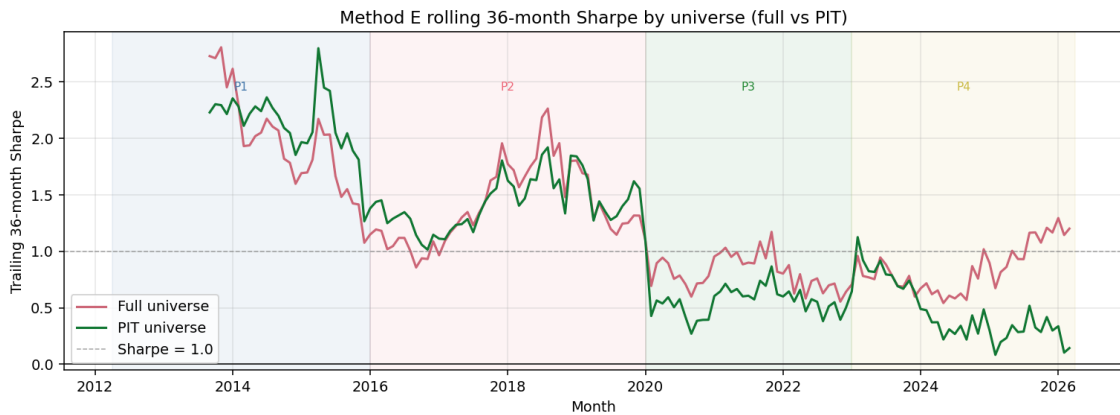


Figure 2: Trailing 36-month Sharpe ratio for Method E on full and PIT universes. The PIT line confirms the sub-period table: a clear monotonic decline from 2018 onwards.

6 Method E alpha survives factor controls (full universe)

Table 4 reports the Fama–French 5-factor + Carhart momentum attribution for Method E’s net-of-cost monthly excess return. The unhedged book, despite long-only construction, has insignificant loadings on every factor. R^2 is 0.027 across all six factors combined — Method E is essentially orthogonal to conventional factor exposure.

Model	α ann	$t(\alpha)$	β_{Mkt}	β_{SMB}	β_{HML}	β_{RMW}	β_{CMA}	β_{MOM}
CAPM	+30.6%	+3.96	−0.15					
FF3	+32.1%	+4.13	−0.22	+0.37	−0.15			
FF5	+31.9%	+4.08	−0.19	+0.37	−0.29	−0.09	+0.36	
FF5 + MOM	+31.2%	+3.94	−0.16	+0.41	−0.24	−0.07	+0.32	+0.12

Table 7: Multi-factor attribution for Method E net excess returns ($n = 167$ months). No factor loading has $|t| > 2$ in any specification. Annualized alpha is stable in the 30–32% range across all models.

7 Alternative explanations

We cannot identify the mechanism that flattens the short side of Cohen–Malloy–Nguyen in this panel, but we can rule out several possibilities:

- **Not sample size.** Decile-level standard errors are well under 5 bps per month, so the observed dispersion among deciles 0–6 is statistically significant and not noise.
- **Not survivorship bias alone.** Survivorship bias should inflate returns roughly uniformly across deciles; it cannot preferentially lift decile 0 above decile 4 unless filing-rewrite firms are disproportionately retained in the index. We have no evidence for this but also no evidence against it — reconstructing PIT membership is still the next step.
- **Not feature collinearity.** The 8-K event score and 10-K cosine similarity are only weakly correlated ($\rho = 0.04$ in the panel), so the ML combiner is not confounding the two signals.

Plausible candidates we cannot falsify from the current data:

- **Regime shift post-GFC.** Cohen et al. study 1994–2017. Our panel starts in 2011, partially overlaps, and extends the out-of-sample window by nine years. A structural shift in how markets price disclosure changes — perhaps driven by faster dissemination or the growth of NLP-driven investors — would show up as exactly this pattern: the long side persists (because it is really a stability/quality tilt) while the short side is arbitrated away.
- **Universe difference.** Cohen et al. use a broader universe; we restrict to S&P 500. Large-cap firms are covered by more analysts and more systematic investors, so the mispricing that supports the short side in their sample may be unavailable in ours.

8 Practical implications

1. **The honest read-out is a decaying signal, not a stable alpha.** PIT Method E delivers Sharpe 0.81 and FF5+MOM alpha +20.6% ($t_{HAC} = +3.07$) over the full 14-year

panel, but breaks down by sub-period: Sharpe 1.4–1.7 in 2012–2019, 0.50 in 2020–2022, and 0.04 in 2023–2026. The 14-year aggregate number is dominated by the early-period working signal. Combined with a deflated Sharpe of 45.6% (selection-adjusted), the PIT-corrected aggregate alpha should be reported with substantial caveats.

2. **Market-neutral reframing is cheap.** Method J costs 8% of Method E’s Sharpe in exchange for exact market neutrality. For capital-constrained long/short books this is the right framing.
3. **Do not build rank-weighted or multi-factor constructions on this signal.** Every such variant failed. Monotonicity is a load-bearing assumption that does not hold here.
4. **The 8-K event score does not subsume the 10-K cosine similarity signal.** Earlier claims that cosine similarity had zero feature importance were based on the final single-window model. Across the 156 walk-forward folds the cosine similarity feature is selected in 68% of folds, with highly variable per-fold importance. No single feature dominates across the panel.

9 Reproducibility

All numbers in this note are produced by the alpha-graph repository at commit 8154c74 (post-cleanup-2026-04-15).

<code>python -m alpha_graph.backtest.improvements</code>	# Table 2
<code>python -m alpha_graph.backtest.pit_universe</code>	# Table 3 (PIT correction)
<code>python -m alpha_graph.backtest.ff_attribution</code>	# Tables 4, 5 (HAC)
<code>python -m alpha_graph.backtest.deflated_sharpe</code>	# Table 6 (DSR)
<code>python -m alpha_graph.backtest.subperiod_stability</code>	# Table 7 + Fig 2
<code>python -m alpha_graph.backtest.realistic_slippage</code>	# cost robustness
<code>python -m alpha_graph.backtest.feature_stability</code>	# feature-importance claim
<code>python scripts/u_shape_plot.py</code>	# Figure 1 and Table 1

References

- [1] Bailey, D. H. and López de Prado, M. (2014). The Deflated Sharpe Ratio: Correcting for Selection Bias, Backtest Overfitting, and Non-Normality. *Journal of Portfolio Management*, 40(5), 94–107.